AI & ML - VITYARTHI PROJECT

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1. Abstract

This project outlines the design and development of an "AI-Powered Library Management System" — a command-line-based Python program that incorporates seven different artificial intelligence and graph algorithms into a fully functional library management system. The library management system has features such as managing books, managing members, managing book issues, calculating overdue fines, and has a range of different artificial intelligence features such as fuzzy search, collaborative filtering for book recommendations, analytics insights, and three different original graph algorithms for literary analysis: "The Ripple" (BFS), "The Rabbit Hole" (DFS), and "The Silk Road" (A* Search). The program does not require any external libraries, API keys, or database servers — it is a native Python program that can be run from the command line.

2. Introduction

Traditional library management systems are data entry and retrieval systems.
They store information and retrieve it in the same manner in which it was entered.
They do not reason, discover, or suggest. This project rethinks the library as a knowledge graph and uses classical AI search techniques to provide a completely new type of interaction with a book collection.
The idea behind this project came to us because we have all had the experience of finding our favorite books not through direct search, but through serendipity, or unexpected connections. A friend recommends a book, an afterword, or a common author of two beloved books. The three literary AI engines in this project were built to find these kinds of connections.

2.1 Objectives

• Develop a fully functional command-line-based library management system
• Implement a minimum of three classical AI/ML algorithms
• Develop a book knowledge graph for the foundation of graph-based search
• Develop The Ripple (BFS), The Rabbit Hole (DFS), and The Silk Road (A*) as literary AI engines
• Develop a fully functional system that requires zero external dependencies
• Develop a fully functional and academically sound system

3. System Overview

3.1 Architecture

The system is divided into three modules:

File	Module	Responsibility
main.py	Interface Layer	All CLI menus, user interaction, feature dispatch
database.py	Data Layer	SQLite CRUD operations, schema setup, data seeding
ai_engine.py	AI/ML Layer	All 7 algorithms: fuzzy search, recommendations, BFS, DFS, A*, insights, overdue analysis

3.2 Database Schema

The system uses SQLite — a file-based relational database built into Python. Three tables store all persistent data:

Table	Columns
books	id, title, author, genre, year, copies, available
members	id, name, email, phone, joined
issued_books	id, member_id, book_id, issue_date, due_date, return_date, fine

The book knowledge graph, which is used in all three of these graph algorithms, is dynamically constructed in-memory from a books table when first needed.
No special database schema is required for any of these algorithms.

4. The Book Knowledge Graph

The basis of all three literary artificial intelligence systems is a directed, weighted adjacency list graph, constructed from the book collection in the library database:
Each book in the collection is a node in the graph, represented by its database ID. Two books are connected with an edge if they are written by the same author (edge weight = 1), or if they are in the same genre (edge weight = 2). The author edges are given a higher weight (stronger edge) since books written by the same author will have a similar creative voice, not merely a similar topic. This graph is constructed in-memory once per session, and BFS, DFS, and A* are performed on this graph.
The construction of this graph is O(n^2) in the number of books in the collection, which is a reasonable size for a library collection of hundreds or thousands of books.

5. AI / ML Algorithms

5.1 Fuzzy Search – Levenshtein Edit Distance

The fuzzy search feature enables users to search for books even if their queries are spelled incorrectly, contain partial words, or use different words.
Implementation: The Levenshtein edit distance is used to determine the similarity between the words in the query and words in the book title, author, and genre.
The books are included in the result list if any word in the book title, author, and genre has a similarity of at least 70%. The result list is ordered according to the confidence score, which varies from 0 to 100.
Time Complexity: O(n * m * k), where n is the number of books, m is the number of words in a book, and k is the number of words in the query.

5.2 Book Recommendations – Collaborative Filtering

The book recommendation system analyzes a member's borrowing history to provide personalized book recommendations.
Implementation: The system calculates a genre affinity score for each genre of
books borrowed by a member, represented as a percentage of total borrows.
Books not yet borrowed by a member are scored according to their genre's affinityscore, and the top five are returned along with a human-readable explanation.
Time Complexity: O(n * m), where n is the number of genres and m is the number of books in a genre.
This is a content-based collaborative filtering method, which does not rely on ratings but on borrowing history. The more a member borrows, the more accurate the recommendations are.

5.3 Insights Dashboard — Analytics & Thresholding

The dashboard displays real-time statistics retrieved from the database. It uses AI to generate natural language comments based on the thresholds. The library utilization is calculated based on the number of books currently issued divided by the total number of books. Thresholds determine whether to recommend buying more books, running a member engagement strategy, or adding diversity to the library. The genre popularity chart is an ASCII art representation.

5.4 Overdue & Fine Analysis — Risk Classification

Each book is assigned a risk classification based on the number of days overdue: LOW (≤14 days), MED (15-30 days), or HIGH (>30 days). The fine charged for each book is ₹2/day. The system displays the total fine that would be charged if all books were returned today. It identifies members who need urgent contact and offers a tip based on average overdue days.

5.5 The Ripple — Breadth-First Search (BFS)

"Every book you love is a stone dropped in still water."
The Ripple is a book discovery system that uses BFS to explore the knowledge graph. It starts with an arbitrary book and displays all related books in expanding waves.
Wave 1 consists of books directly related to the starting book.
Wave 2 consists of books related to Wave 1, and Wave 3 consists of books related to Wave 2.

Algorithm details:
• Data structure: A queue implemented as a deque to allow O(1) enqueue and dequeue operations
• Use of a set to keep track of visited nodes to avoid cycles
• Each discovered node is marked with its 'hop distance' and type of 'bond'
• Time complexity: O(V + E), where V is number of books, E is number of edges connecting authors/genres
Innovation: The algorithm's output is referred to as 'waves' rather than 'levels', where each 'wave' has a different type of connection:
📚 same author
📚 same genre
📚 chain
This makes the algorithm's output meaningful to a user, not just a programmer.

5.6 The Rabbit Hole — Depth-First Search (DFS, Recursive)

"You didn't plan to read this many books. But one led to another."
The Rabbit Hole algorithm uses a recursive DFS to follow one chain of readings as deep as possible before backtracking. Author links have a weight of 1, and genre links have a weight of 2, so the algorithm dives as deep as possible into an author's entire body of work before 'drifting' off into different genres. The algorithm's output is visualized as an indentation: each level of indentation represents one link deeper into the hole.

Algorithm details:
• Recursive function with visited set to prevent cycles
• Neighbour list is sorted by weight: author links (1) before genre links (2)
• Cap on maximum depth set to 5 to prevent runaway recursion with large collections
• Time complexity: O(V + E)
Innovation:

The output provides statistics on the depth achieved. The metaphor of 'falling' into the world of reading is taken literally with the indentation structure.

5.7 The Silk Road — A* Search

"Ancient traders didn't teleport from Rome to China. They traveled through connected cities."
The Silk Road algorithm will find the optimal path between two selected books. It does this using the A* search algorithm. It is possible to calculate the cost of the path using the actual cost g(n), combined with the admissible cost function h(n).

Algorithm details:
• Priority queue: heapq library for a min-heap priority queue
• g(n) function: cumulative cost of reaching node n – author link: 1, genre link: 2
• h(n) function: proximity heuristic for genres – pre-defined table of distances between genres
  (e.g., Fiction-Dystopia: 1, Fantasy-Science Fiction: 1, Self-Help-Business: 2, etc.)
• The heuristic function h(n) is admissible – it never overestimates the path length
• best_g dictionary stores the minimum known cost to each node, avoiding re-expansion
• Time complexity: O(ElogV) in the average case
Innovation: The final result provides a rating of the smoothness of the journey depending on the total cost, and each waypoint is marked by the type of terrain. The "no path found" condition is handled nicely with a message and a suggestion to add more books to the collection.

6. Results & Testing

The system has been tested on a pre-defined data set of 20 books in 10 genres and 5 members. Key results:

Test Case	Input	Result
Fuzzy Search — typo	"harey poter"	✅Harry Potter found (66%)
Fuzzy Search — genre misspelling	"dystopiaa"	✅1984 & Brave New World (71%)
Recommendations — Member 1	Self-Help borrow history	✅Self-Help & Fiction books returned
The Ripple — Book 6 (Harry Potter)	Fantasy genre, BFS depth 3	✅The Hobbit returned in Wave 1
The Rabbit Hole — Book 9 (Atomic Habits)	Self-Help genre, DFS depth 4	✅Deep Work discovered (same genre)
The Silk Road — Book 1 to Book 2	Fiction to Fiction, A*	✅Direct path, cost = 2
Overdue Analysis	2 overdue books in test data	✅Fines calculated, LOW risk

7. Conclusion

This project proves that effective AI integration is possible without the use of large language models, cloud APIs, or libraries. The meaningful AI integration is achieved by applying classical AI algorithms such as BFS, DFS, A* search, edit distance, and collaborative filtering on the book collection knowledge graph provided by the library management system. The three AI engines for the project, The Ripple, The Rabbit Hole, and The Silk Road, are not just algorithm implementations but also provide meaningful functionality for the reader/user. The names provided for the AI engines are also relevant to the reader/user, as opposed to the algorithm implementation itself. This is another important feature that must be taken into consideration while creating an AI system. The project is complete and can be executed from the command line with one command, meeting all the requirements set for the project.

7.1 Future Enhancements

• Weighted graph with borrow frequency edges for more personalisation
• Natural Language Querying with the help of a local LLM
• Exporting the reading path from The Silk Road as a shareable reading list
• Web-based User Interface layer on top of the existing AI Engine implementation
• Multiple library federation with The Silk Road covering multiple library collections

8.Output Screenshot

```
📚 AI-Powered Library Management System
The Ripple ⁂ The Rabbit Hole ⁂ The Silk Road ⁂ More

MAIN MENU

📖 BOOK MANAGEMENT
 1.  Add a Book
 2.  View All Books
 3.  Search Books (exact)
 4.  Delete a Book

👤 MEMBER MANAGEMENT
 5.  Add a Member
 6.  View All Members

🔄 TRANSACTIONS
 7.  Issue a Book
 8.  Return a Book
 9.  View Issued Books

🤖 AI / ML FEATURES
10.  🔍 Smart Fuzzy Search
11.  💡 Book Recommendations
12.  📊 Insights Dashboard
13.  ⚠️  Overdue & Fine Analysis

📜 LITERARY AI ENGINES
14.  🌊 The Ripple        [BFS]
15.  🐇 The Rabbit Hole   [DFS]
16.  🗺️  The Silk Road    [A*]

 0.  Exit

Enter your choice:
```

```
── SEARCH BOOKS ──────────────────────────

Title / Author / Genre: harry poter
No results.

Press Enter to continue...
```

```
── 🔍 SMART FUZZY SEARCH (Levenshtein) ──────────────

Type anything — handles typos, partial words, synonyms.

Your search: harry puttar

ID    Title                              Author                 Genre            Score
──────────────────────────────────────────────────────────────────────────────────────
6     Harry Potter and the Sorcerer's Stone J.K. Rowling           Fantasy          40%

Press Enter to continue...
```

```
── ALL MEMBERS ──────────────────────────

ID    Name                     Email                        Phone          Joined
──────────────────────────────────────────────────────────────────────────────────
1     Aarav Sharma             aarav@email.com              9876543210     2026-03-30
5     Amit Kumar               amit@email.com               9765432109     2026-03-30
2     Priya Patel              priya@email.com              9123456789     2026-03-30
3     Rohan Verma              rohan@email.com              9988776655     2026-03-30
4     Sneha Gupta              sneha@email.com              9871234567     2026-03-30

Press Enter to continue...
```

```
── ISSUED BOOKS ──────────────────────────

IssID  Member               Title                          Issue        Due          Status
──────────────────────────────────────────────────────────────────────────────────────────
8      Sneha Gupta          Sapiens                        2026-03-11   2026-03-25   ⚠️ OVERDUE
9      Amit Kumar           Rich Dad Poor Dad              2026-03-13   2026-03-27   ⚠️ OVERDUE
5      Priya Patel          The Hobbit                     2026-03-19   2026-04-02   ✅ OK
3      Aarav Sharma         Deep Work                      2026-03-21   2026-04-04   ✅ OK
6      Rohan Verma          Introduction to Algorithms     2026-03-23   2026-04-06   ✅ OK

Press Enter to continue...
```

```
⚠️  OVERDUE BOOKS & FINE ANALYSIS

Member              Title                       Due         Days   Fine     Risk
──────────────────────────────────────────────────────────────────────────────
Sneha Gupta         Sapiens                     2026-03-25  5      ₹10.00   🟢 LOW
Amit Kumar          Rich Dad Poor Dad           2026-03-27  3      ₹6.00    🟢 LOW
──────────────────────────────────────────────────────────────────────────────
Total fines: ₹16.00

🤖 AI ANALYSIS
• Avg overdue: 4.0 days
• 💡 Keep reminders going.

Press Enter to continue...
```

```
📊 AI LIBRARY INSIGHTS

Total Books      : 20
Available        : 61
Currently Issued : 5
Total Members    : 5
Fines Collected  : ₹0.00
Utilization Rate : 25.0%

🤖 Utilization is healthy.

📚 GENRE POPULARITY
──────────────────────────────────────────
Self-Help          (2)
Fantasy            (2)
Computer Sci.      (2)
History            (1)
Finance            (1)
Fiction            (1)

🤖 Top genre: 'Self-Help'

🏆 MOST BORROWED BOOKS
──────────────────────────────────────────
1. The Hobbit (1 borrows)
2. Clean Code (1 borrows)
3. Introduction to Algorithms (1 borrows)
4. Rich Dad Poor Dad (1 borrows)
5. Deep Work (1 borrows)

⚠️  2 OVERDUE BOOK(S)
• Sapiens - Sneha Gupta (5d overdue)
• Rich Dad Poor Dad - Amit Kumar (3d overdue)

Press Enter to continue...
```

```
⚠️  OVERDUE BOOKS & FINE ANALYSIS

Member              Title                       Due          Days   Fine      Risk
Sneha Gupta         Sapiens                     2026-03-25   5      ₹10.00    🟢 LOW
Amit Kumar          Rich Dad Poor Dad           2026-03-27   3      ₹6.00     🟢 LOW

Total fines: ₹16.00

🤖 AI ANALYSIS
• Avg overdue: 4.0 days
• 💡 Keep reminders going.

Press Enter to continue...
```

```
✦ Wave 1  — Books directly linked (same author or genre)
✦ Wave 2  — Books linked to those
✦ Wave 3  — The outermost reach of influence

Algorithm: Breadth-First Search (BFS)
```

ID	Title	Author	Genre
3	1984	George Orwell	Dystopia
9	Atomic Habits	James Clear	Self-Help
4	Brave New World	Aldous Huxley	Dystopia
14	Clean Code	Robert C. Martin	Computer Sci.
11	Deep Work	Cal Newport	Self-Help
16	Dune	Frank Herbert	Science Fiction
6	Harry Potter and the Sorcerer's Stone	J.K. Rowling	Fantasy
13	Introduction to Algorithms	Thomas Cormen	Computer Sci.
18	Pride and Prejudice	Jane Austen	Romance
12	Rich Dad Poor Dad	Robert Kiyosaki	Finance
8	Sapiens	Yuval Noah Harari	History
5	The Alchemist	Paulo Coelho	Philosophy
7	The Da Vinci Code	Dan Brown	Thriller
1	The Great Gatsby	F. Scott Fitzgerald	Fiction
17	The Hobbit	J.R.R. Tolkien	Fantasy
10	The Lean Startup	Eric Ries	Business
15	The Pragmatic Programmer	Andrew Hunt	Computer Sci.
19	The Psychology of Money	Morgan Housel	Finance
20	Thinking, Fast and Slow	Daniel Kahneman	Psychology
2	To Kill a Mockingbird	Harper Lee	Fiction

```
Drop your stone — enter Book ID: 1

🪨 Your stone: "The Great Gatsby" by F. Scott Fitzgerald
   Genre: Fiction

Dropping into the library waters...

🌊 FIRST WAVE  —  Directly connected — same author or genre

│  [ 2] To Kill a Mockingbird          Fiction        ← 🧬 same genre
│   1 book(s) in this wave

🌊 The Ripple reached 1 books across 1 waves.
💡 The closer the wave, the stronger the reading connection.
```

```
Algorithm: Depth-First Search (DFS) — Recursive
──────────────────────────────────────────────────────────────────────────────

ID    Title                                Author                 Genre
──────────────────────────────────────────────────────────────────────────────
3     1984                                 George Orwell          Dystopia
9     Atomic Habits                        James Clear            Self-Help
4     Brave New World                      Aldous Huxley          Dystopia
14    Clean Code                           Robert C. Martin       Computer Sci.
11    Deep Work                            Cal Newport            Self-Help
16    Dune                                 Frank Herbert          Science Fiction
6     Harry Potter and the Sorcerer's Stone J.K. Rowling               Fantasy
13    Introduction to Algorithms           Thomas Cormen          Computer Sci.
18    Pride and Prejudice                  Jane Austen            Romance
12    Rich Dad Poor Dad                    Robert Kiyosaki        Finance
8     Sapiens                              Yuval Noah Harari      History
5     The Alchemist                        Paulo Coelho           Philosophy
7     The Da Vinci Code                    Dan Brown              Thriller
1     The Great Gatsby                     F. Scott Fitzgerald    Fiction
17    The Hobbit                           J.R.R. Tolkien         Fantasy
10    The Lean Startup                     Eric Ries              Business
15    The Pragmatic Programmer             Andrew Hunt            Computer Sci.
19    The Psychology of Money              Morgan Housel          Finance
20    Thinking, Fast and Slow              Daniel Kahneman        Psychology
2     To Kill a Mockingbird                Harper Lee             Fiction

Choose where to fall — enter Book ID: 4

🕳️ Falling into: "Brave New World" by Aldous Huxley
   Genre: Dystopia

THE HOLE  (each level = one link deeper)
──────────────────────────────────────────────────────────────────────────────

🕳️  [4] "Brave New World"
     by Aldous Huxley  (Dystopia)
     ↓ falling...

   │  📖 [3] "1984"
   │       by George Orwell  (Dystopia)
   │       🧬 Same Genre

──────────────────────────────────────────────────────────────────────────────
🕳️ You fell 1 level(s) deep.
   ✍️   Author links followed : 0
   🧬 Genre links followed  : 1
```

```
18    Pride and Prejudice              Jane Austen            Romance
12    Rich Dad Poor Dad                Robert Kiyosaki        Finance
8     Sapiens                          Yuval Noah Harari      History
5     The Alchemist                    Paulo Coelho           Philosophy
7     The Da Vinci Code                Dan Brown              Thriller
1     The Great Gatsby                 F. Scott Fitzgerald    Fiction
17    The Hobbit                       J.R.R. Tolkien         Fantasy
10    The Lean Startup                 Eric Ries              Business
15    The Pragmatic Programmer         Andrew Hunt            Computer Sci.
19    The Psychology of Money          Morgan Housel          Finance
20    Thinking, Fast and Slow          Daniel Kahneman        Psychology
2     To Kill a Mockingbird            Harper Lee             Fiction

Where do you want to arrive? Book ID: 13

🏛️  ORIGIN      : "Clean Code" by Robert C. Martin  [Computer Sci.]
🚩  DESTINATION : "Introduction to Algorithms" by Thomas Cormen  [Computer Sci.]

Charting the route...

✅ Route found!  Waypoints: 2   Travel cost: 2   Smoothness: ⭐⭐⭐ Very Smooth

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🏛️  ORIGIN       [14] "Clean Code"
             by Robert C. Martin
             Genre: Computer Sci.

🚩  DESTINATION  [13] "Introduction to Algorithms"
             by Thomas Cormen
             Genre: Computer Sci.
             🌉 Genre bridge — a gentle shift in landscape

═══════════════════════════════════════════════════════════

🗺️  THE SILK ROAD SUMMARY
──────────────────────────────────────────
Total waypoints  : 2 book(s)
Travel cost      : 2 (lower = smoother)
Journey smoothness: ⭐⭐⭐ Very Smooth

💡 Read these books in order — each one is a stepping
   stone that makes the next feel natural, not jarring.
   A* guarantees this is the least-friction route possible.
```

```
── ALL BOOKS ──────────────────────────

ID    Title                            Author                Genre            Yr     Cop  Avail
──────────────────────────────────────────────────────────────────────────────────────────────
3     1984                             George Orwell         Dystopia         1949   4    4
9     Atomic Habits                    James Clear           Self-Help        2018   5    5
4     Brave New World                  Aldous Huxley         Dystopia         1932   2    2
14    Clean Code                       Robert C. Martin      Computer Sci.    2008   2    2
11    Deep Work                        Cal Newport           Self-Help        2016   2    1
16    Dune                             Frank Herbert         Science Fiction  1965   3    3
6     Harry Potter and the Sorcerer's Stone J.K. Rowling          Fantasy          1997   6    6
13    Introduction to Algorithms       Thomas Cormen         Computer Sci.    2009   3    2
18    Pride and Prejudice              Jane Austen           Romance          1813   3    3
12    Rich Dad Poor Dad                Robert Kiyosaki       Finance          1997   4    3
8     Sapiens                          Yuval Noah Harari     History          2011   4    3
5     The Alchemist                    Paulo Coelho          Philosophy       1988   5    5
7     The Da Vinci Code                Dan Brown             Thriller         2003   3    3
1     The Great Gatsby                 F. Scott Fitzgerald   Fiction          1925   3    3
17    The Hobbit                       J.R.R. Tolkien        Fantasy          1937   4    3
10    The Lean Startup                 Eric Ries             Business         2011   3    3
15    The Pragmatic Programmer         Andrew Hunt           Computer Sci.    1999   2    2
19    The Psychology of Money          Morgan Housel         Finance          2020   4    4
20    Thinking, Fast and Slow          Daniel Kahneman       Psychology       2011   2    2
2     To Kill a Mockingbird            Harper Lee            Fiction          1960   2    2

Press Enter to continue...
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╔═══════════════════════════════════════════════════╗
   📖   T H E   S I L K   R O A D                   ║
   The most elegant path between two literary worlds
╚═══════════════════════════════════════════════════╝

You are here. You want to be there.
But the jump feels too large — different genres, different worlds.

The Silk Road charts the smoothest overland route between
two books, using connecting titles as waypoints. Like the
ancient trade routes, it avoids unnecessary detours and
finds the path of least literary friction.

✦ Each waypoint = a book that bridges the gap
✦ Author links  = short, smooth passage (cost: 1)
✦ Genre links   = a gentle shift in terrain (cost: 2)
✦ The route is optimal — A* guarantees no shorter path exists

Algorithm: A* Search with genre-distance heuristic
──────────────────────────────────────────────────────

── YOUR ORIGIN ──────────────────────────────────────

ID    Title                            Author                Genre
──────────────────────────────────────────────────────────────────
3     1984                             George Orwell         Dystopia
9     Atomic Habits                    James Clear           Self-Help
4     Brave New World                  Aldous Huxley         Dystopia
14    Clean Code                       Robert C. Martin      Computer Sci.
11    Deep Work                        Cal Newport           Self-Help
16    Dune                             Frank Herbert         Science Fiction
6     Harry Potter and the Sorcerer's Stone J.K. Rowling          Fantasy
13    Introduction to Algorithms       Thomas Cormen         Computer Sci.
18    Pride and Prejudice              Jane Austen           Romance
12    Rich Dad Poor Dad                Robert Kiyosaki       Finance
8     Sapiens                          Yuval Noah Harari     History
5     The Alchemist                    Paulo Coelho          Philosophy
7     The Da Vinci Code                Dan Brown             Thriller
1     The Great Gatsby                 F. Scott Fitzgerald   Fiction
17    The Hobbit                       J.R.R. Tolkien        Fantasy
10    The Lean Startup                 Eric Ries             Business
15    The Pragmatic Programmer         Andrew Hunt           Computer Sci.
19    The Psychology of Money          Morgan Housel         Finance
20    Thinking, Fast and Slow          Daniel Kahneman       Psychology
2     To Kill a Mockingbird            Harper Lee            Fiction
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